

Q1.

(a) Show that $\frac{1}{5 \cosh x - 3 \sinh x} = \frac{e^x}{m + e^{2x}}$, where m is an integer.

(3)

(b) Use the substitution $u = e^x$ to show that

$$\int_0^{\ln 2} \frac{1}{5 \cosh x - 3 \sinh x} dx = \frac{\pi}{8} - \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right)$$

(5)

(Total 8 marks)

Q2.

(a) Given that $u = \cosh^2 x$, show that $\frac{du}{dx} = \sinh 2x$.

(2)

(b) Hence show that

$$\int_0^1 \frac{\sinh 2x}{1 + \cosh^4 x} dx = \tan^{-1}(\cosh^2 1) - \frac{\pi}{4}$$

(5)

(Total 7 marks)

Q3.

By using the substitution $u = x - 2$, or otherwise, find the exact value of

$$\int_{-1}^5 \frac{dx}{\sqrt{32 + 4x - x^2}}$$

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(Total 5 marks)